LittleGenius

AI Learning Platform for Kids

MERN Stack Edition (MongoDB, Express.js, React, Node.js)

1. Project Overview

LittleGenius is an interactive educational platform designed to make learning more engaging and accessible for children aged 6 to 12.

The platform allows teachers to create educational courses, organize lessons, manage classes, and monitor student progress. Students can access their assigned courses, explore educational content, complete interactive activities, and track their learning achievements.

Preparing exercises and educational activities adapted to different ages and learning levels can be time-consuming for teachers. LittleGenius addresses this challenge by integrating artificial intelligence to help teachers generate quizzes, exercises, and explanations based on their lesson content.

Teachers remain responsible for reviewing, modifying, and approving AI-generated content before making it available to students.

The project aims to combine three main objectives:

- **Simplify educational content creation:** Provide teachers with tools to organize courses and prepare learning activities with AI assistance.
- **Encourage interactive learning:** Offer children an engaging learning environment with activities adapted to their age and educational level.
- **Support learning progress:** Allow students to track their achievements while giving teachers visibility into student performance and difficulties.

The final application should provide a complete learning experience, from course creation and AI-assisted activity generation to student participation and progress monitoring.

2. Users and Roles

LittleGenius supports two main user roles.

Teacher

Teachers are responsible for creating and managing educational content and organizing their students. They should be able to:

- Create and manage their own classes.
- Add students to their classes and manage their accounts.
- Create courses, chapters, sections, and learning activities.
- Use AI to generate educational activities from lesson content.
- Review and modify AI-generated content before publishing.
- Assign courses to their classes.
- Monitor student progress and activity results.

Each teacher manages their own classes, courses, and students independently.

Student

Students access a dedicated learning environment where they can:

- Log in using their individual accounts.
- View courses assigned to their classes.
- Navigate through lessons and educational content.
- Complete interactive learning activities.
- Receive feedback on their answers.
- Track their course progress and learning achievements.

Students should only have access to their assigned educational content and personal learning records.

3. Functional Requirements

3.1 Must Have: Required Features

The following features are essential for LittleGenius to deliver its primary educational value.

A. Authentication and User Management

The platform must provide secure authentication and appropriate access permissions.

- Teachers and students must be able to log in using email and password.
- Teachers should be able to create and manage their own profiles.
- Teachers must be able to create classes and add or remove students.
- Students must have individual accounts linked to their teacher's classes.
- Users should only access the resources and functionality permitted by their roles.
- Teachers should only manage students, classes, and educational content that belong to them.

B. Course and Content Management

Teachers must be able to create structured educational courses. Each course should follow a clear hierarchy: **Course > Chapters > Sections**.

- **Course:** Represents a subject or learning topic, such as Mathematics.
- **Chapter:** Organizes a course into major learning units.
- **Section:** Contains lesson content and associated learning activities.

The platform should allow teachers to:

- Create, edit, organize, and delete courses, chapters, and sections.
- Add educational explanations, examples, and instructions to sections.
- Create interactive activities associated with specific sections.
- Prepare courses as drafts before publishing them.
- Assign published courses to one or more of their classes.
- Update existing course content when necessary.

Students should only see published courses assigned to their classes. The course structure should support coherent learning paths rather than collections of unrelated exercises.

C. Interactive Learning Activities

LittleGenius must support three types of learning activities.

Activity Type	Description
Multiple-Choice Quiz	Students select the correct answer from several available options.
Matching	Students match related elements, such as words and definitions or questions and answers.
Fill in the Blanks	Students complete sentences or explanations by providing missing words or expressions.

For each activity:

- Teachers should be able to create activities manually or generate them using AI.
- Students should be able to interact with activities and submit answers.
- The application must automatically evaluate answers when applicable.
- Students should receive clear feedback after completing an activity.
- Results must be saved to support learning progress tracking.
- Activities should remain understandable and suitable for children aged 6 to 12.

D. AI-Assisted Activity Generation

Teachers must be able to use artificial intelligence to generate educational activities directly from their lesson content. The platform should allow teachers to:

- Select an existing section or provide educational content.
- Choose the activity type they want to generate.
- Request AI-generated questions, answers, and explanations.
- Review the generated activity before accepting it.
- Edit incorrect, unclear, or unsuitable content.
- Regenerate an activity if the result is unsatisfactory.
- Save approved activities as part of the selected section.

Generated activities must be compatible with the platform's interactive learning system and should behave like manually created activities.

AI-generated content must never be published automatically without teacher approval.

E. Student Learning Experience

The platform should provide a dedicated learning environment designed for children. Students should be able to:

- Access their assigned courses from a personal learning area.
- Navigate through chapters and sections.

- Read lesson explanations and examples.
- Complete interactive quizzes and exercises.
- Receive feedback after submitting answers.
- Identify completed activities and sections.
- Continue learning from their previously recorded progress.

The interface should prioritize simplicity, readability, clear navigation, and engaging interactions appropriate for younger users.

F. Student Progress Tracking

LittleGenius must track student learning activity and provide relevant progress information.

- The application must record completed activities and student results.
- Students should be able to view their progress within each course.
- Teachers should be able to monitor the progress of students in their classes.
- Teachers should be able to review activity scores and completion status.
- The platform should help teachers identify students who experience difficulties with particular lessons or activities.
- Progress information must remain available between learning sessions.

3.2 Should Have: Recommended Features

These features strengthen the educational experience and improve the usefulness of the platform.

A. Detailed Learning Feedback

Students should receive explanations when answering incorrectly, helping them understand their mistakes. Teachers should be able to provide explanations manually or use AI assistance to generate them.

B. Activity Difficulty Levels

Teachers should be able to select the intended difficulty of an activity. AI-generated exercises should consider the selected difficulty and the age group of the students.

C. Teacher Progress Overview

Teachers should have access to a clear summary of class performance, including:

- Course completion rates.
- Average activity results.
- Student participation and progress.
- Activities or topics presenting recurring difficulties.

D. Activity Retries

Students should be able to repeat selected learning activities for additional practice. The application should distinguish between previous attempts and updated results to maintain meaningful progress records.

3.3 Could Have: Optional Features

The following improvements offer opportunities for creativity and additional technical exploration.

- **Points and Badges:** Reward students for completing lessons, achieving good results, or reaching learning milestones.
- **Learning Streaks:** Encourage consistent participation by recognizing regular learning activity.
- **AI Practice Recommendations:** Suggest additional exercises based on topics where students are struggling.
- **Visual Activities:** Introduce images or other educational media into learning exercises.
- **Personalized Learning:** Adapt selected activities or recommendations to individual student performance.

Students are encouraged to propose other improvements that strengthen LittleGenius's educational value and overall learning experience.

4. AI Integration

4.1 Purpose

The main purpose of AI in LittleGenius is to simplify the creation of educational activities while helping teachers prepare varied and age-appropriate learning material.

Instead of manually writing every quiz or exercise, teachers should be able to transform their lesson content into interactive activities using AI assistance. AI should support the teacher's work without replacing their responsibility for educational accuracy and content quality.

4.2 Expected AI Workflow

The AI functionality should support the following workflow:

- **1. Lesson Preparation:** A teacher creates a section and provides the educational content, such as an explanation of fractions or basic grammar rules.
- **2. Activity Selection:** The teacher chooses the activity type and, when applicable, the intended difficulty level.
- **3. AI Generation:** The application sends the relevant lesson content and generation instructions to the AI service.
- **4. Generated Result:** The AI produces an activity containing appropriate questions, answers, and supporting explanations when relevant.
- **5. Teacher Validation:** The teacher reviews the generated content, modifies it when necessary, and decides whether to approve or regenerate it.
- **6. Publication:** The approved activity becomes part of the lesson and can be accessed by students once the course is published.
- **7. Student Participation:** Students complete the activity, and the application evaluates their responses and records their results.

4.3 Expected AI Capabilities

The AI integration should be capable of:

- Understanding educational content provided by teachers.
- Generating questions that remain relevant to the lesson topic.
- Producing activities appropriate for children aged 6 to 12.
- Supporting quizzes, matching exercises, and fill-in-the-blanks.
- Providing correct answers and explanations when needed.
- Adapting generated content to a requested difficulty level.
- Returning structured results that can be validated and used by the application.

4.4 Technical Considerations

The AI functionality must be implemented within the Express.js backend, as a dedicated AI service module (for example, a separate layer that builds the prompt from the lesson content, calls the chosen AI API or model, and validates the structured response). No separate AI backend is required.

The solution should account for several important considerations:

- AI-generated questions may contain factual mistakes or ambiguous answers.
- Generated content must be validated before being stored or used.
- Teachers must always have the opportunity to review and modify generated activities.
- The application should handle invalid AI responses and generation failures appropriately.
- Teachers should still be able to create educational activities manually when AI generation is unavailable.
- AI functionality should be achievable using free APIs, open-source models, or suitable free-tier services.

Students are responsible for researching and selecting appropriate models, APIs, prompting strategies, and generation techniques.

5. Technology Stack and Deployment

5.1 Required Technology Stack (MERN)

The project must be built exclusively with the MERN stack. No other framework, language, or database is required or expected.

Component	Technology
Database	MongoDB (with Mongoose as the ODM)
Backend	Node.js with Express.js (main application logic and AI integration)
Frontend	React.js (functional components and hooks, with React Router for navigation)
AI Integration	AI API or model integrated through the Express.js backend
Version Control	Git and GitHub

MongoDB is recommended for LittleGenius because its flexible document structure is suitable for courses containing chapters, sections, and different types of educational activities.

The Express.js backend is responsible for the main application logic, authentication, authorization, data management, and AI-related operations. The React frontend consumes the backend REST API.

Students are responsible for choosing the application's architecture, designing the database, and organizing the AI module within the backend.

5.2 Deployment

The completed application must be deployed and accessible online. All components, including the frontend, Express.js backend, MongoDB database, and AI integration, must function together in the deployed environment.

Suggested deployment services include:

Component	Suggested Service
React Frontend	Vercel or Netlify
Express.js Backend (including AI integration)	Railway or Render
MongoDB Database	MongoDB Atlas

These services are recommendations rather than mandatory choices. Students may select alternative hosting providers, preferably using free tiers or available free resources.

6. Expected Deliverables

The final project submission should include:

- A functional full-stack LittleGenius application built with the MERN stack.
- Authentication and role-based authorization for teachers and students.
- Class and student management.
- Course, chapter, section, and activity management.
- Interactive learning activities with automatic evaluation.
- Student progress tracking and teacher monitoring.
- Functional AI-assisted activity generation.
- Persistent MongoDB database integration.
- Source code available in a GitHub repository.
- A README containing the project overview, installation steps, configuration, and deployment instructions.
- Sample courses, classes, students, and learning activities for demonstration.
- A live deployed application accessible online.
- A final presentation and demonstration of the main workflows and technical decisions.

Example End-to-End Scenario

- **1. Class and Student Setup:** A primary school teacher creates a class for nine-year-old students and adds their individual accounts.
- **2. Course and Lesson Preparation:** The teacher creates a Mathematics course titled Introduction to Fractions, organizes it into chapters, and prepares a section explaining halves and quarters with simple examples.
- **3. AI Activity Generation:** Using the lesson content, the teacher requests a multiple-choice quiz and a matching activity. The AI generates questions, answers, and relevant explanations.
- **4. Teacher Review and Publication:** The teacher notices that one question is too difficult for the age group, edits it, approves both activities, and publishes the course to the class.
- **5. Student Learning:** Adam logs in to his student account, opens the assigned course, reads the lesson, and completes both activities.
- **6. Automatic Evaluation and Progress:** LittleGenius evaluates Adam's answers, provides feedback, and records his results. He understands halves but struggles with questions about quarters. His progress remains available when he returns.
- **7. Teacher Progress Review:** The teacher checks class participation and activity results, notices that several students struggled with quarters, and prepares an additional explanation and practice activity for the topic.

This scenario demonstrates the complete learning workflow, from course creation and AI-assisted activity generation to teacher approval, student participation, automatic evaluation, and teaching decisions informed by recorded progress.

7. Evaluation Criteria

The project will be evaluated according to the following criteria:

Criterion	Description
Functional Completeness	Correct implementation of the required features and main educational workflows.
Frontend and User Experience	Interface quality, responsiveness, ease of use, and suitability for teachers and children.
Backend and Database	Quality of API development, data management, authentication, authorization, and business logic.
AI Integration	Relevance, reliability, and practical usefulness of AI-generated educational activities.
Code Quality	Organization, readability, maintainability, and appropriate technical decisions.
Deployment	Accessibility and correct integration of all application components.
Team Collaboration	Effective collaboration, Git usage, individual contributions, and understanding of the solution.
Final Demonstration	Ability to demonstrate complete workflows, explain technical choices, and present a functional application.

The expected outcome is a coherent, functional, and engaging educational platform that demonstrates the ability to combine full-stack MERN development and artificial intelligence into a practical solution for teachers and young learners.